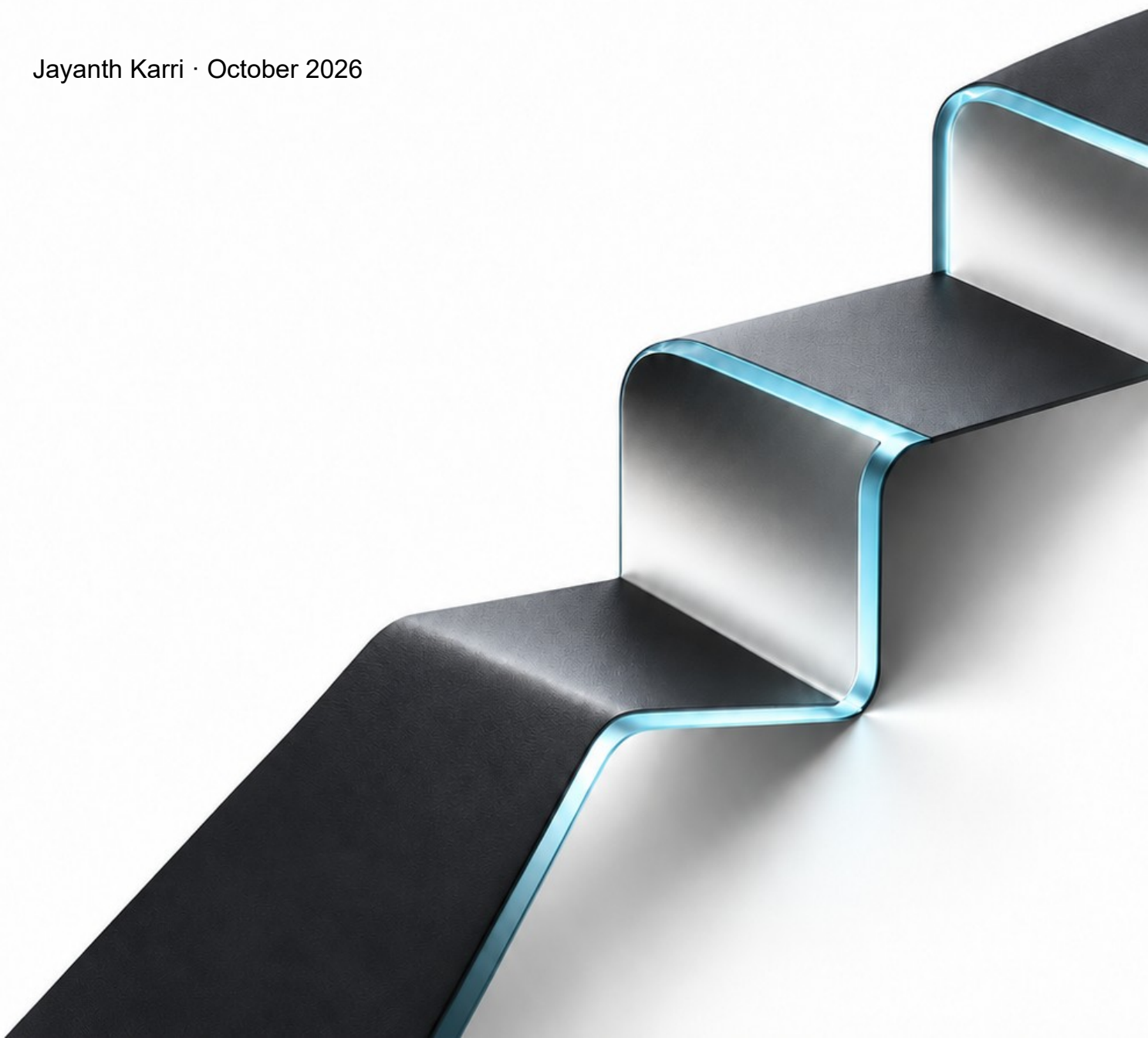
Beyond the Alert

Jayanth SOC Automation Lab

Connecting Detection Evidence to Analyst Investigation

Jayanth Karri · October 2026



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Jayanth SOC Automation Lab — Connecting Detection Evidence to Analyst Investigation

Whitepaper | Prepared by Jayanth Karri | Wording revision 21 October 2026

Executive summary

In a security operations center (SOC), an alert is the beginning of an investigation. An analyst still has to establish what happened, collect relevant context, prepare a case, and decide whether further action is justified. This lab explores how connected tools can assist that repeated work while leaving investigation and response decisions with the analyst.

Beyond the Alert brings together Windows telemetry, Splunk detection, n8n automation, GPT-4.1 Mini, reputation research, DFIR-IRIS case management and Slack notifications. The goal is a consistent investigation package with an alert summary, indicators, supporting context, recommended checks and evidence links. This paper examines the lab design and reported study; the implementation paper documents the current worker and verification. [1-4]

Reported lab evaluation

The supplied comma-separated values (CSV) file contains 15 runs across three configurations. Its recorded durations average 209.00 seconds for manual triage, 130.00 seconds for AI without enrichment tools, and 38.11 seconds for the integrated configuration. The last figure is 81.76% lower than the manual mean. These are recalculated summaries of reported lab measurements, not new measurements or production benchmarks. Unequal scenario coverage and timing limitations affect their interpretation. [5]

Lab artifacts and configuration

The collection illustrates detection searches, workflow logic, case views, notifications and reported results. Its earlier source export is inactive with five disconnected webhook entry points; reconnect those routes before replay. The current implementation has a separate connected export and local checks, described in the implementation paper. [1-3]

How to read this paper

Pages 2-6 explain the problem and system. Pages 7-8 walk through an account-change example. Pages 9-10 explain evaluation and results. Pages 11-12 set out practical improvements, conclusions, and references. Technical terms are defined as they appear.

Source context. This paper uses Final Copy of Artifacts and the cited earlier repository snapshot. Selected screenshots include presentation updates to names and displayed dates, documented in the source records. October 2026 is the presentation timeline. This artifact-based study and the current synthetic implementation checks have separate evidence scopes; service-connected validation is the next milestone. [1,6]

The problem is repeated triage work

Consider an analyst receiving an alert about repeated failed logins. The alert provides a clue, but it does not explain whether someone mistyped a password, a service has stale credentials, or an attacker is trying multiple passwords. The analyst needs the account, source address, target computer, timing, and surrounding activity before reaching a conclusion.



Figure 1. Conceptual illustration generated for this paper. Left: repeated switching between tools. Right: organized findings support analyst review.

Where the repeated handling occurs

A manual investigation involves opening a detection, checking related logs, researching indicators, writing notes, creating a case and notifying a colleague. Each handoff requires careful preservation of host, source-event time and case context. Connecting those steps addresses repeated handling; this study examines the design and reported lab sample.

The proposed solution

The proposed workflow passes the alert between tools rather than requiring the analyst to carry every field manually. Splunk identifies a matching pattern. n8n coordinates processing. The model summarizes the input and requests relevant reputation checks. JavaScript prepares structured requests for IRIS and formats an analyst message. This makes the information easier to review together. [2-4]

Enrichment means adding useful context to an existing alert, such as a public IP reputation result or a file-hash report. It does not turn an indicator into a confirmed incident. A private address is still an investigation lead even when external reputation services cannot assess it.

The analyst remains responsible

The useful security outcome is a better starting point for investigation. The analyst validates events, checks authorization, resolves contradictory fields and selects a response. The referenced workflow prepares records and notifications; endpoint isolation and firewall changes require separate authorized response tooling. [2]

Architecture connects evidence and investigation

The architecture starts with endpoint records and ends with information an analyst can investigate. A security information and event management system, or SIEM, collects and searches logs. Workflow orchestration coordinates tasks across services. Figure 2 shows the intended component relationships. [1,4]

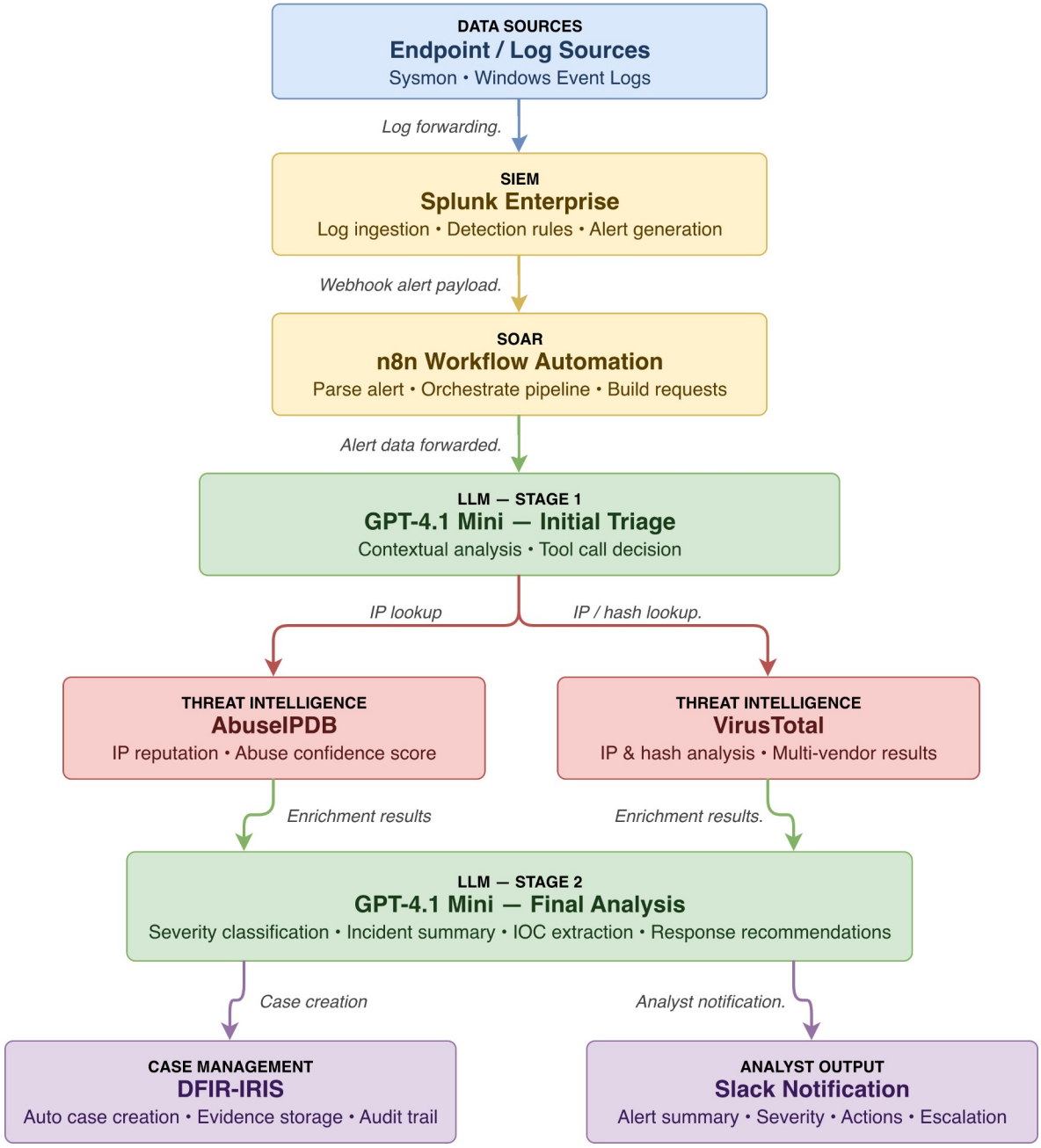


Figure 2. Lab architecture, artifact 07. The conceptual map shows initial reasoning and final analysis. The earlier export uses one model node with connected tools; the current worker enriches indicators before one validated advisory assessment. LLM means large language model. [1-2]

Referenced sequence. After model processing, the earlier export prepares an IRIS alert, escalates it into a case and checks severity before Slack delivery. High/Critical gates Slack, while case creation precedes that condition. The current sequence is documented in the implementation paper. [2]

The lab separates service roles

Five virtual machines separate endpoint, simulation, detection, orchestration and case-management roles on VMnet8 NAT subnet 192.168.106.0/24. Network address translation (NAT) provides outbound connectivity. Apply explicit access and transport controls when configuring this lab. [4]

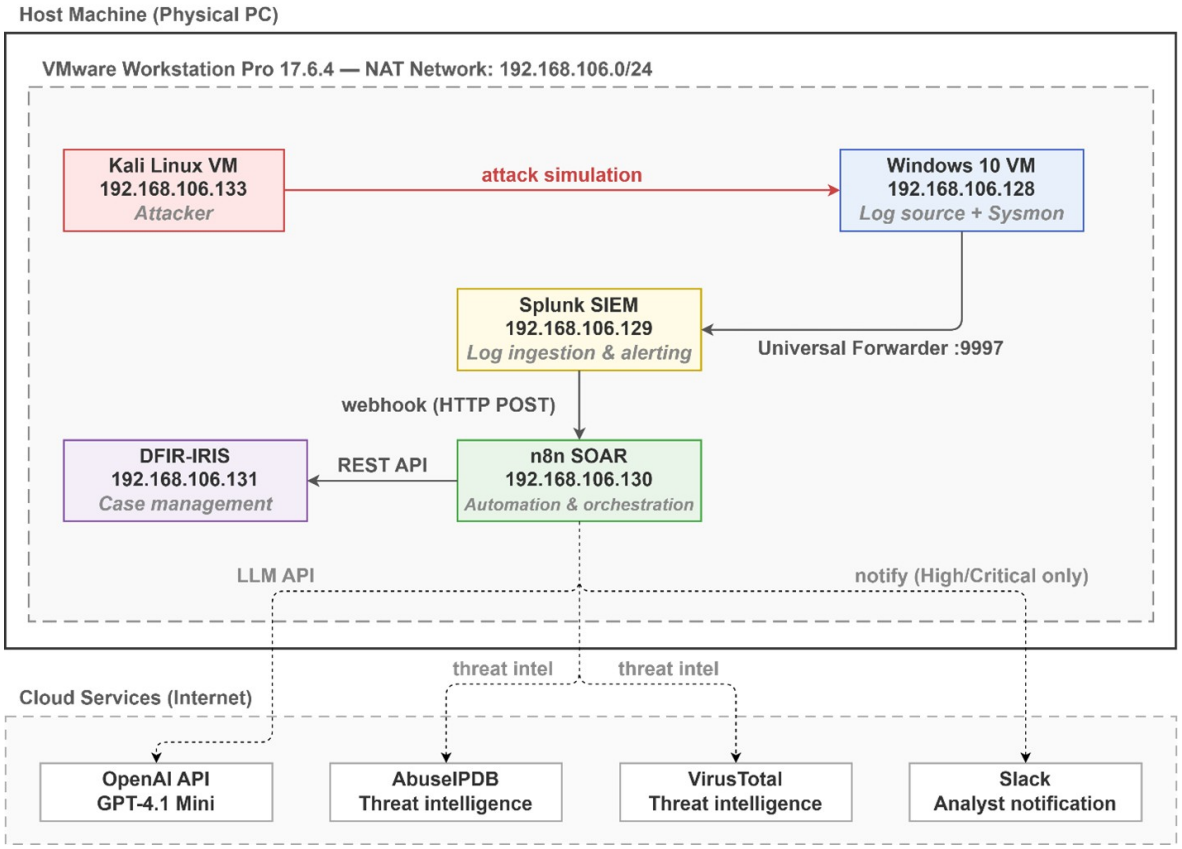


Figure 3. Supplied topology, artifact 13. Follow Kali to Windows, Windows forwarding to Splunk, and Splunk webhooks to n8n. The lower paths identify external application programming interfaces (APIs). [1,4]

Machine	Address	Documented role	RAM
Windows 10	192.168.106.128	Target and telemetry	4 GB
Splunk on Ubuntu	192.168.106.129	Ingestion and detection	4 GB
n8n on Ubuntu	192.168.106.130	Workflow orchestration	4 GB
IRIS on Ubuntu	192.168.106.131	Investigation records	4 GB
Kali Linux	192.168.106.133	Controlled simulation	2 GB

The documented endpoint forwards logs to Splunk on TCP 9997. Splunk sends webhooks to n8n on port 5678; n8n communicates with the IRIS HTTPS API, model, reputation providers and Slack. Artifacts 04-05 show VM inventory and NAT settings. Validate service health through runtime checks and retained execution records. [1,4]

Capacity and setup. The listed VMs allocate 18 GB before host overhead; size the host above those simultaneous allocations rather than relying on the source notes' 16 GB minimum. Record service versions, credentials, forwarding inputs and tested endpoints during reproduction. [4]

Detection turns events into investigation leads

An event is a recorded activity; a detection applies logic to events; an alert asks for review. Splunk Search Processing Language, or SPL, expresses those conditions. The supplied queries cover five scenarios. They document intended coverage rather than detection completeness. [3]

Scenario	Evidence and configured condition
Failed logins	Security event 4625; group by computer, user, and source IP; count at least 3 within the selected window.
File hash	Sysmon event 1; extract SHA256 and return a recent process event for reputation review.
Network scanning	Firewall DROP records; at least 3 distinct selected ports or 5 drops per source and destination.
Account changes	Security events 4720 or 4732; account creation or matching Administrators membership change.
PowerShell	Process and command-line patterns such as encoding, download functions, or execution-policy bypass.

```
index="Jayanth - Project" EventCode=4625
| eval user=coalesce(user, Account_Name, TargetUserName)
| eval src_ip=coalesce(src_ip, Source_Network_Address, IpAddress)
| stats count min(_time) as _time by ComputerName user src_ip
| where count >= 3
| table _time ComputerName user src_ip count
```

Figure 4. Enlarged query excerpt, artifact 08. Inspect EventCode 4625, grouping by computer/user/source IP and count >= 3. The full source retains its matching result and provenance label. [1,3]

The full artifact reports count 11 for DESKTOP-JKLABPC1, user Jayanth, and private source 192.168.106.1. That result exceeds the query threshold but does not establish malicious intent or a scheduled webhook firing. Check the time window and alert schedule separately. Microsoft defines event 4625 as a failed logon. [3,7]

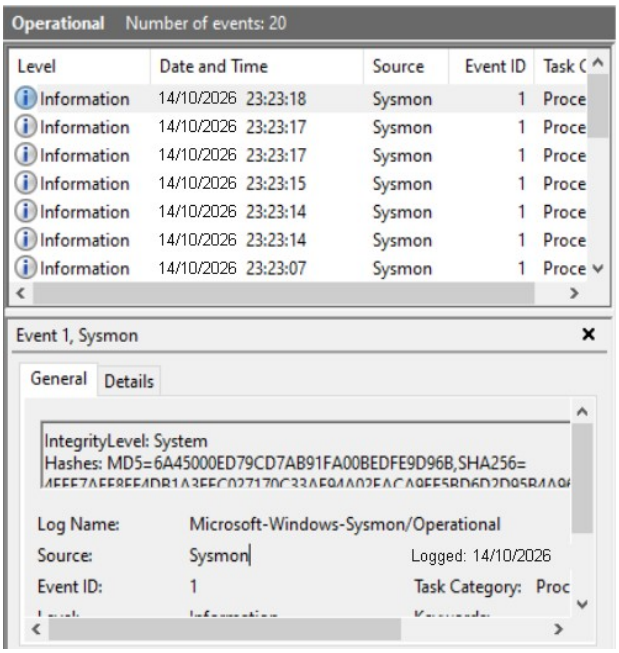


Figure 5. Enlarged process-telemetry excerpt, artifact 09. Event ID 1 and process details identify the source. Retrieve the complete event for any hash lookup because the displayed hash is partly clipped. [1,8]

Sysmon records process context, including command lines and configured hashes; it does not decide whether the process is malicious. Likewise, an encoded PowerShell flag can justify review without proving compromise. [3,8]

Workflow logic for cases and notifications

n8n accepts alerts through webhook endpoints. Figure 6 illustrates the earlier lab workflow. The following walkthrough describes its referenced export; the maintained n8n route and JavaScript worker have a separate current implementation record. [2,9]

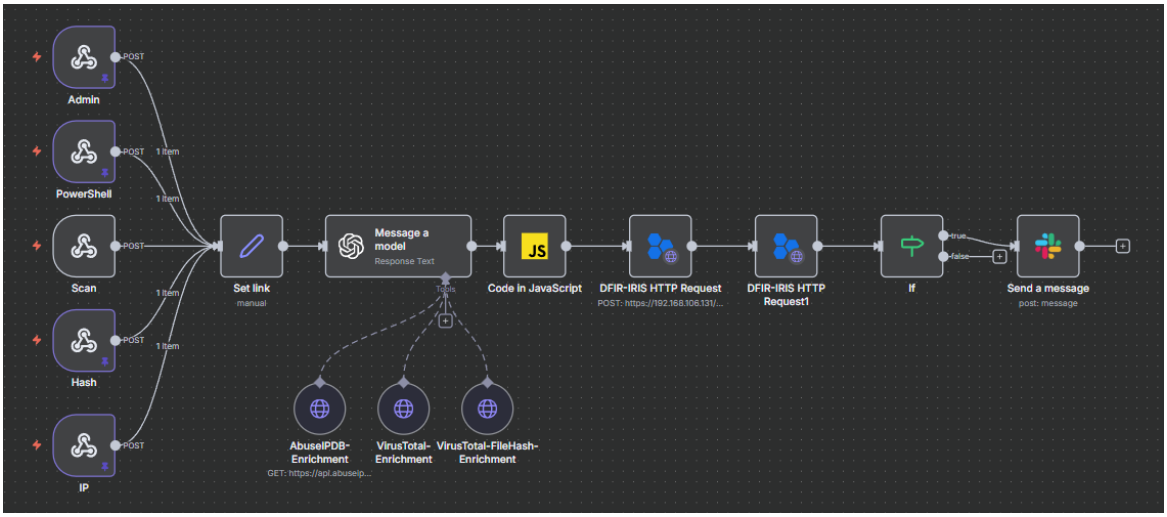


Figure 6. Workflow canvas, artifact 01. Five alert types feed analysis and attached enrichment tools. Workflow configuration and pinned samples document the automation logic and example data flow; the earlier export requires webhook reconnection before replay. [1-2]

Follow the processing path

1. Receive and retain context. IP, Hash, Scan, PowerShell and Admin identify the five input types. Set link retains incoming fields and the source-results link. Reconnect each earlier-export entry point and validate its downstream route during reproduction. [2]

2. Interpret and enrich. The model is instructed to return JSON, a structured field format. Public IPs can trigger AbuseIPDB and optional VirusTotal checks; SHA256 values use the file-hash tool. Private IPs skip external reputation. Account-creation alerts take a priority path with a High minimum and no reputation lookups. This severity floor is a policy instruction, not external corroboration. Prompt instructions do not prove runtime enforcement. [2]

3. Build requests. JavaScript parses the response, calculates a risk level, builds tags and notes, and prepares IRIS request bodies. The export then calls /alerts/add followed by /alerts/escalate/{alert_id}. Its indicator and asset import arrays are empty, even though notes may contain indicator text. Structured indicator of compromise (IOC) import therefore needs separate verification. [2]

4. Notify for review. The If node passes High or Critical results to the Slack node, which uses a saved API credential and selected channel. The code calculates channel labels, but these labels alone do not establish dynamic channel routing. [2]

Reproducing the referenced workflow

The earlier export has active set to false and five empty webhook connection arrays. Configure credentials/URLs, map the index, reconnect routes and deliberately select or clear pinned samples before a controlled run. The maintained export instead has connected success/failure routes and remains inactive until configured and service-validated. n8n distinguishes test and production webhook URLs. [2,9]

Worked example account and group changes

The investigation question is whether a new account and administrator-group change on DESKTOP-JKLABPC1 were authorized. Event 4720 records user creation; event 4732 records addition to a security-enabled local group. A group change requires checking the actual group and member, not treating every 4732 as an administrator change. [3,10-11]

```
index="Jayanth - Project" source="WinEventLog:Security" (EventCode=4720 OR EventCode=4732)
| eval action=case(
    EventCode=4720, "User Account Created",
    EventCode=4732, "Added to Local Administrators Group"
)
| eval is_admin_add=if(EventCode=4732 AND like(_raw,"%Administrators%"),1,0)
| where EventCode=4720 OR is_admin_add=1
```

target_account	Raw_Splunk_time	actions	domain	host	initiating_user	computer	hits
Jayanth	1792102677.251	Added to Local Administrators Group User Account Created	DESKTOP-JKLABPC1	DESKTOP-JKLABPC1	socadmin	DESKTOP-JKLABPC1	2
socadmin	1792102667.788	User Account Created	DESKTOP-JKLABPC1	DESKTOP-JKLABPC1	socadmin	DESKTOP-JKLABPC1	1

Figure 7. Enlarged excerpts of artifact 12: query conditions above and selected result columns below. The full source retains omitted columns and provenance. Screenshot names and dates follow the presentation history described on page 1. [1,3]

Read the result before accepting the label

The Jayanth row lists both User Account Created and Added to Local Administrators Group, with two hits. The initiating-user field shows socadmin. A second row concerns socadmin account creation. These fields tell the analyst which identities and host to examine; the names do not establish intent. The query assigns HIGH, so that label is not independent corroboration of compromise.

Validate the underlying events

Open the source events and confirm target and subject security identifiers, group identity, event record IDs, computer, and time zone. Check whether the initiating account had an approved administrative task. Review nearby successful logons and process activity where available. Aggregation can combine multiple actions, so preserve event-level records rather than relying only on the grouped row. [3,10-11]

The rule uses an English Administrators text match and groups by target account. These choices need validation against field extraction, localized group names, and multi-host activity. The collection supplies screenshots and samples, not the native event records required to independently reconstruct this change. [1,3]

The case supports an analyst decision

The next artifacts show case and notification context: severity, host, ownership, summary and source links. Use them to follow an investigation handoff; the displayed case remains Open. Analysts validate the evidence and authorize response and closure. [1-2]

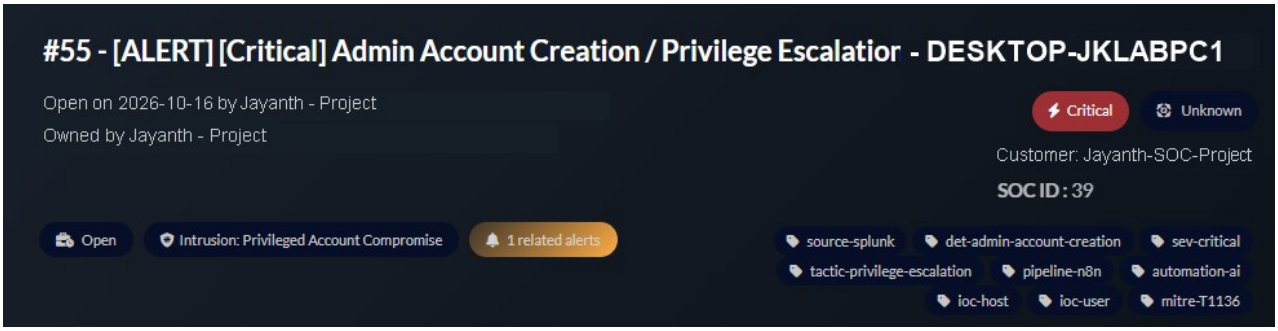


Figure 8. IRIS case-header excerpt, artifact 03. Case 55 is Open and Critical, references the lab host and carries investigation tags. The full source includes the escalation summary and source label. [1]

This artifact-based walkthrough requires identity reconciliation: the case summary names eviladmin, while Figure 7 shows socadmin. Validate the initiating user against source records before joining these views into one execution trace. A correlated run retains source-event references, alert ID, workflow execution ID and downstream case ID together.

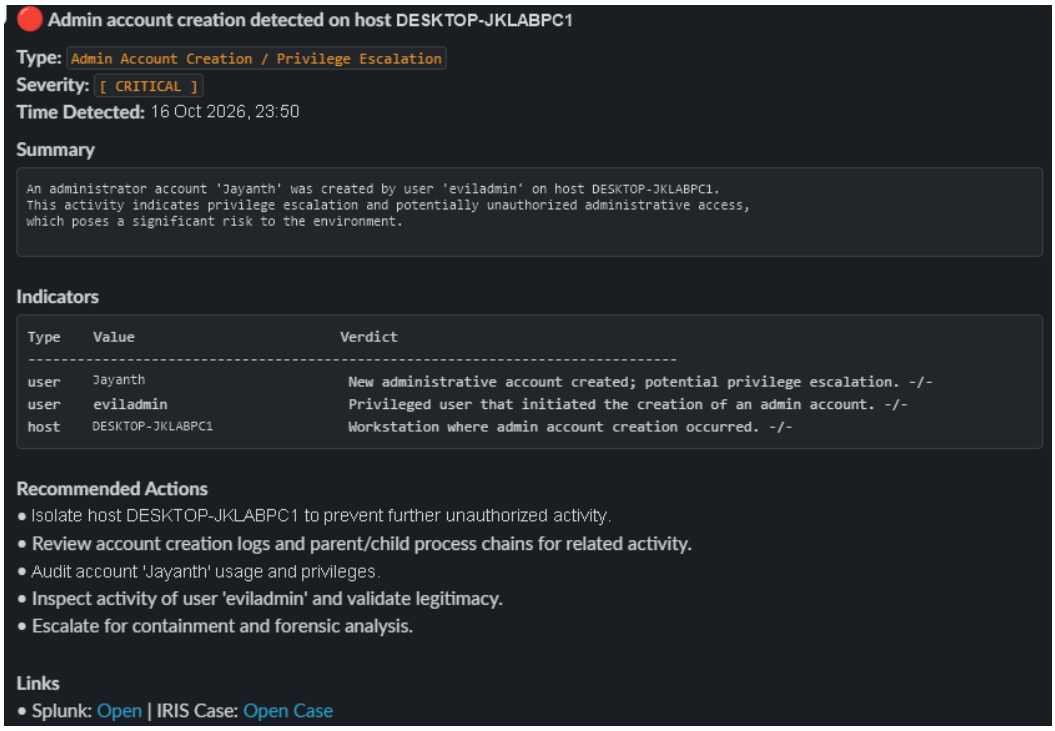


Figure 9. Notification and response-guidance excerpt, artifact 02. The message names eviladmin and recommends investigation and isolation. The crop omits interface margins and the original footer; source presentation history is documented on page 1. [1]

The analyst checks authorization, resolves the identity mismatch and assesses related activity before choosing a response. The notification provides guidance. Execution and successful containment require separate authorized tooling and a recorded outcome. [1-2]

Evaluation compares three configurations

The supplied evaluation asks whether coordinated processing and relevant external context help triage compared with manual handling and AI alone. Its unit of observation is one reported test run. The CSV contains 15 rows across five scenario families, but the three configurations cover different mixes. [5]

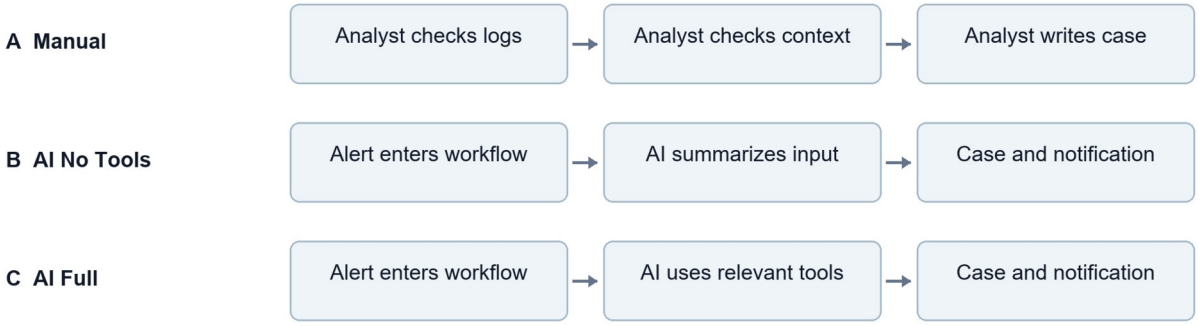


Figure 10. Explanatory comparison drawn for this paper from the supplied method. It describes the configurations, not a newly executed experiment. [2,5]

Configuration	Runs	Scenario coverage
A Manual	5	One run per scenario family.
B AI No Tools	3	Failed logins, file hash, and scanning.
C AI Full	7	All five families; two login and two hash runs.

Define the measurements first

Triage time is the recorded triage_time_s value in seconds. The method describes a boundary from a Splunk alert through final notification or manual documentation, while also listing detection overhead. Analyst steps count discrete human actions needed to complete triage. Action quality and context quality are single-reviewer scores on 1-5 rubrics. Means in this paper use every stored row in each configuration. [5]

Interpret the timing record carefully

The retained dashboard and review notes describe recorded durations for runs 1, 3, 6 and 7 as totals including detection and pipeline-processing overhead. This paper uses the original triage_time_s values and preserves their source explanation. Future evaluation should capture stage timestamps automatically under one run ID. [5]

Keep the lab substitutions visible

Because the VMs use private addresses, public indicators were substituted in selected payloads to exercise reputation lookups. Those tests show reported behavior for substituted inputs, not evidence that an external attacker reached the lab. File-hash scenarios include a benign and a known-malicious example. The account and PowerShell scenarios were not included in configuration B. This design cannot isolate the causal effect of enrichment across all five families. [3,5]

Reported results favor integrated triage

Recalculation reproduces the supplied means below. Compared with Manual, the AI Full mean duration is 81.76% lower and its mean analyst-step count is 41.56% lower. These are descriptive differences between small groups with unequal scenario mixes, not guaranteed improvements for another SOC. [5]

Configuration	Runs	Mean time	Mean steps	Action / 5	Context / 5
Manual	5	209.00 s	6.60	3.00	3.60
AI No Tools	3	130.00 s	3.00	4.00	3.00
AI Full	7	38.11 s	3.86	4.57	4.29

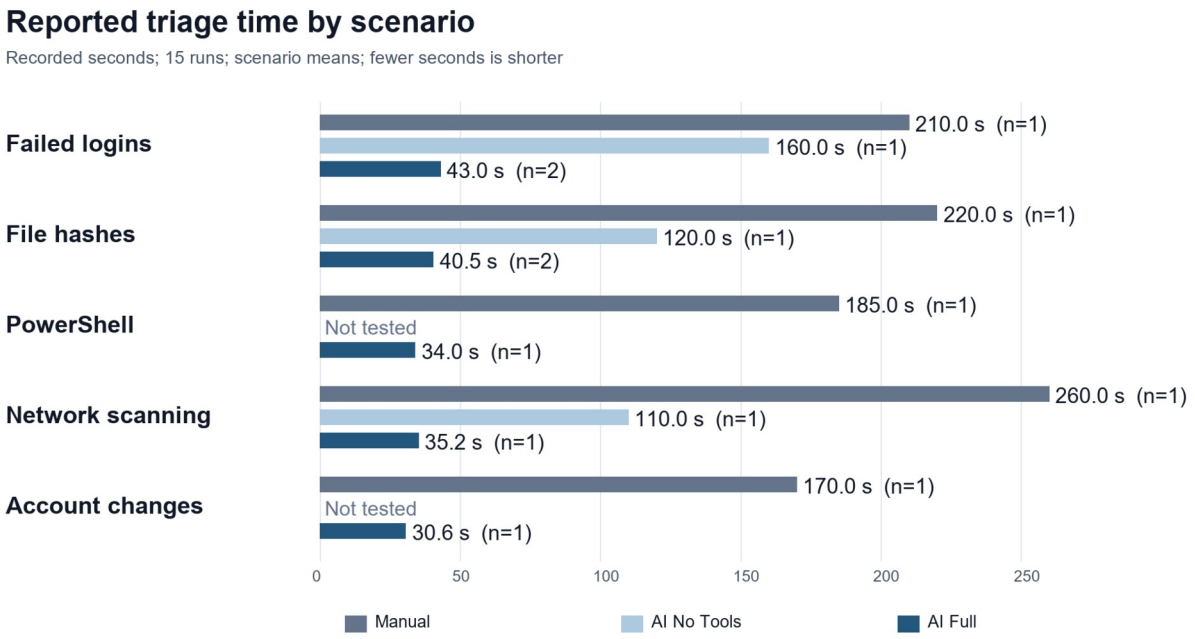


Figure 11. Scenario means recalculated from evaluation-results.csv. Each bar includes its sample count. No Tools has no account-change or PowerShell observation. Shared scenarios are still not matched repeated trials. [5]

What the comparisons suggest

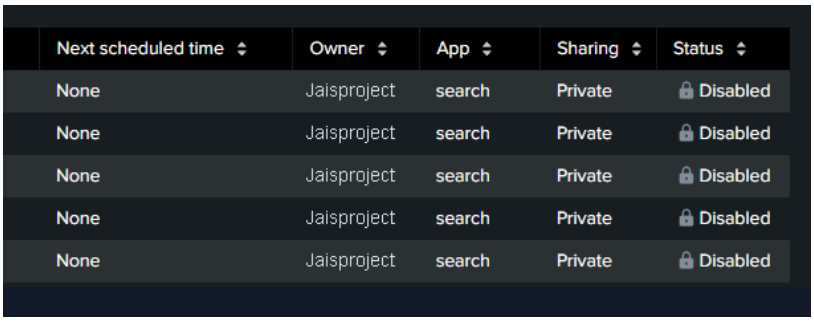
The largest recorded manual duration is scanning at 260 seconds, compared with 35.2 seconds for AI Full. Failed-login means are 210 versus 43 seconds; hash means are 220 versus 40.5 seconds. Configuration B uses fewer reported human steps than C, but that does not establish better investigation quality: it omits external tool context, and its scenario mix differs. [5]

The quality scores favor AI Full in this sample. They are subjective ratings from one researcher, not independent proof that recommendations were correct. Tool availability, scenario selection, and test conditions could explain part of the differences. No production throughput, cost per case, false-positive rate, or containment success was measured by this review. [5]

Artifact 14 is a supplied normalized radar chart. It remains reported evaluation material; this paper uses the CSV-derived values instead of reproducing its normalized axes. A published 100% classification claim cannot be recalculated from the CSV because separate ground-truth and predicted-class fields are absent. [1,5]

Operational value and validation priorities

Beyond the Alert connects detection context with investigation preparation and an analyst handoff. The lab study supports the design discussion. The current implementation adds authenticated routing, input/output checks and reproducible examples; the next milestone is service-connected validation with linked runtime records. [1-5]



Next scheduled time	Owner	App	Sharing	Status
None	Jaisproject	search	Private	Disabled
None	Jaisproject	search	Private	Disabled
None	Jaisproject	search	Private	Disabled
None	Jaisproject	search	Private	Disabled
None	Jaisproject	search	Private	Disabled

Figure 12. Rule-inventory excerpt, artifact 11. Five saved rules show Disabled with no next scheduled time. The collection describes disabling after testing; deployment requires activation and baseline checks. [1,3]

Reproduce the earlier export

For the earlier export, reconnect five webhook inputs, configure endpoints/credentials and add the error path. Confirm forwarding and field extraction with a benign test before enabling the intended rule. Retain source events, mapped alert, workflow execution, enrichment result, case ID and notification time under one run ID. [2-4]

Earlier export security and current controls

The earlier source export lacks webhook authentication and permits untrusted IRIS certificates. These findings apply to that referenced configuration. The current implementation uses authenticated intake and worker requests, trusted TLS requirements, bounded transport and duplicate handling. Validate these controls with deployed services before increasing volume. [2,9]

The earlier parser can continue with an empty object after invalid JSON and substitute processing time for source time. The maintained worker instead validates input, advisory schema, identities and quotations, and preserves observation timestamps. Untrusted log content remains evidence; analysts review semantic correctness and authorize actions. See the implementation paper for the current contract. [2]

Measure investigation quality as well as speed

Repeat matched scenarios across all applicable configurations, including benign and ambiguous examples. Record stage timestamps automatically, API failures, cost, ground truth, predicted labels, and reviewer corrections. Use a second reviewer for quality scoring. Test private-IP bypass, missing hash results, malformed output, repeated alerts, and failed case creation. [5]

The open-case queue illustrates prioritization and investigation organization. Closure requires an analyst verdict, supporting evidence and verified response outcomes. The lifecycle diagram describes that analyst process; authorized containment uses separate response tooling. [1-2]

Conclusion glossary and references

Beyond the Alert explains how detection context, indicator research and case preparation support SOC investigation. Its reported lab comparison motivates the design, while the accompanying implementation provides reproducible synthetic examples and checked processing logic. The next milestone is a correlated service-connected run, followed by matched evaluation and documented analyst outcomes.

Essential terms

Term	Meaning here
SOC / SIEM	Security operations center / system for collecting and searching security events.
SOAR	Security orchestration automation and response; n8n coordinates this lab workflow.
IOC / SHA256	Indicator of compromise or investigation lead / a file-content fingerprint.
Enrichment / JSON	Adding external context / a structured data format.
SPL / webhook	Splunk query language / HTTP endpoint used to receive an alert.

Evidence and attribution

This paper was prepared from the supplied adapted artifact collection and the cited repository snapshot. The repository is credited as the implementation reference; Jayanth Karri is the whitepaper author. Names and displayed event dates were adjusted in selected demonstration images; source files and review records provide provenance. Crops are labeled excerpts. Figure 1 is AI-generated; Figures 10-11 are newly drawn explanatory and analytical visuals. No generated image is offered as lab evidence. [1,6]

References

[1] Final Copy of Artifacts Collection README, Artifact Bible, and numbered figures.

[2] Workflow export Supplied workflows/pipeline-c.json; pinned source counterpart.

[3] Detection rules and scenarios Supplied SPL and attack-scenario notes.

[4] Lab environment Supplied amended lab notes; pinned source counterpart.

[5] Evaluation records CSV source counterpart; supplied timing review in the artifact collection.

[6] Provenance records Image edit regions and sample timestamp updates.

[7] Microsoft event 4625 Failed-logon event reference.

[8] Microsoft Sysmon Process telemetry and event definitions.

[9] n8n Webhook documentation Webhook URLs, activation, and authentication.

[10] Microsoft event 4720 User-account creation reference.

[11] Microsoft event 4732 Local security-group membership reference.

Earlier source snapshot: 41866043d4581ed85d4e921d26bb5e42625af299. Collection references open the portable artifact guide; external citations retain the original source context. Current implementation behavior is documented separately. References checked 19 October 2026; wording revised 21 October 2026.